

Ayurvedic Medicine Identification and Health Management App

![Project Banner](project_banner.jpg) < (<https://github.com/IT21813184-kavi/AyushathraResearch-Mobile-/blob/component/it21584718/TongueDiagnosticIdentification/assets/Ayushasthra%20main%20logo.png?raw=true>) >

Project Overview

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This mobile application integrates **AI and Ayurveda** to make traditional medicine more **accessible**, **personalized**, and **digitally enhanced**. The app empowers users to:

- Identify **Ayurvedic medicinal plants**
- Diagnose **tongue and skin-related diseases**
- Monitor **mental well-being**
- Receive **customized home remedies , lifestyle suggestions** rooted in Ayurvedic practices

The app aims to bridge the gap between **holistic traditional care** and **modern AI-powered healthcare**, addressing the increasing demand for natural and personalized wellness solutions.

Features

AI-Powered Tongue Diagnostics

- Detects visible patterns or abnormalities on the human tongue
- Classifies tongue conditions into **Normal**, **Mild**, and **Severe** stages using image analysis
- Provides **health insights**, **home remedies**, and **nutritional recommendations**
- Supports **preventive care** and self-awareness

Skin Disease Detection & Ayurvedic Guidance

- Classifies skin conditions into **Normal**, **Moderate**, and **Severe** stages using image analysis
- Offers personalized **lifestyle recommendation**, and **dietary plans**

Medicinal Plant Identification

- AI-based plant image recognition for **Ayurvedic herbs and leaves**
- Displays **usage guidelines**, **benefits**, and **associated disease treatments**

4. Mental Well-being Assessment via Facial Emotion Analysis

- Uses emotion detection to monitor **emotional state trends**
- Suggests **meditation**, **breathing exercises**, and **Ayurvedic mood enhancers**
- Builds a stress assessment model based on **emotion fluctuation patterns**

Technology Stack

- **Frontend**: Flutter (Dart)
- **Backend**: Python Flask
- **Machine Learning**: TensorFlow, Keras, OpenCV, Scikit-learn
- **Database**: Firebase
- **Image Collection & Training**: Google Colab, Jupyter Notebook
- **Version Control**: Git, GitHub
- **Version Control**: VS Code

AI / Machine Learning Models

- **Tongue Disease Detection** - CNN model trained on annotated tongue images and using Mobile V-Net
- **Skin Disease Classification** - Image classifier with stage-based output using CNN (Custom CNN and Inception CNN)
- **Facial Emotion Recognition** - EfficientNetB0: A lightweight and optimized model, designed to achieve high accuracy with fewer parameters and computations
- **Plant Identification** - VGG16: A classic deep CNN with 16 layers (13 convolutional + 3 fully connected)

App Architecture

![[System Diagram](system_diagram.jpg)] < (<https://github.com/IT21813184-kavi/AyushathraResearch-Mobile-/blob/master/assets/sys.jpg?raw=true>) >

The system combines:

- **Frontend (Flutter)** for user interactions
- **Backend (Flask)** for model processing and API responses
- **Cloud (Firebase)** for real-time data sync
- **AI Models** for medical insights and plant recognition
- **Tools**: VS Code, Jupyter Notebook, Google Colab, Git, GitHub

Installation

Prerequisites

- Flutter SDK: [Install Guide](<https://flutter.dev/docs/get-started/install>)
- Python (3.8+): [Download](<https://www.python.org/>)
- Flask: `pip install flask`
- Firebase Setup
- Git: [Install Git](<https://git-scm.com/>)